

Digital microscopy

EVOS M7000 Imaging System

The Invitrogen™ EVOS™ M7000 Imaging System—technical specifications

The EVOS M7000 Imaging System is an automated digital inverted microscope for 4-color fluorescence, transmitted light, and color imaging.

Optics	
Imaging modes	Fluorescence, brightfield, color brightfield, and phase contrast.
Imaging methods	Single color, multicolor, area scan with montage or tile-stitch, time-lapse, Z-stacking, and movie capture.
Optical system	Infinity-corrected optical system; RMS-threaded objectives with 45 mm parfocal distance.
Illumination	Adjustable-intensity LED cubes (>50,000-hour lifetime) with integrated hard-coated filters.
Light cube capacity	5-position chamber for 4 fluorescent LED cubes and brightfield imaging.
Light cubes	Selection from 13 Invitrogen™ EVOS™ LED cubes, including: • DAPI (357/447 nm) • GFP (482/524 nm) • RFP (542/593 nm) • Texas Red (585/628 nm) • Cy®5 (635/692 nm) Motorized fluorescent LED cube interchange mechanism. Custom LED cubes available on request.
Objective capacity	5-position automated turret
Objectives	Selection of more than 30 high-quality, long working distance (LWD) and coverslip-corrected (CC) objectives; magnification from 1.25x to 100x.
Condenser	60 mm LWD condenser; 4-position turret with a clear aperture and 3 phase annuli.
Focus mechanism	Automated focus with sub-micron (0.15 µm) resolution and single-step accuracy.
Cameras	High-sensitivity 3.2 MP monochrome CMOS camera (2,048 x 1,536 pixels) with 3.45 µm pixel resolution. High-sensitivity 3.2 MP color CMOS camera (2,048 x 1,536 pixels) with 3.45 µm pixel resolution.
Captured images	16-bit RAW monochrome: TIFF, PNG (12-bit dynamic range) 8-bit color: TIFF, PNG, JPG Movies and time-lapse: AVI, WMV
LCD display	23-inch high-resolution touchscreen color monitor, fully controllable via mouse; 1,920 x 1,080 resolution.
Mechanics	
Stage control	Motorized
x-axis and y-axis control	Motorized; travel range: 120 x 80 mm with sub-micrometer resolution.
z-axis control	Automated, motorized z-axis software control.
Inserts	Wide selection of drop-in inserts for vessel holders and lock-down holders to keep samples in place during long scans.

Vessels	
Compatibility	Microscope and chamber slides Hemocytometers 6-, 12-, 24-, 48-, 96-, and 384-well microplates 35, 50, 60, and 100 mm petri dishes T-25, T-75, and T-175 flasks Custom vessel configurations available on request.
Automation	
	Automated scans with multiple options for automation routines.
Software and PC	
Integrated onboard operating software	Autofocus Cell counting Confluence measurements Transfection efficiency measurements Field overlap adjustments Batch analysis Annotation tool Stage speed settings 10x Genomics™ slide support
Invitrogen™ Celleste™ Image Analysis Software (optional)	Functions for counting, segmenting, classifying, and analyzing complex images. Preconfigured analysis templates for common applications and an icon-based, wizard-driven workflow. Modules for 2D and 3D deconvolution, 3D rendering, 3D visualization, and 3D analysis.
Image saving	Images can be saved on the embedded hard drive, an external USB device, or a local network.
Computer	External Dell™ 3650 PC with an eleventh generation Intel™ Core™ i7-11700 processor; NVIDIA™ Quadro™ RTX™ 4000 graphics card compatible with Windows™ 10 and designed to operate with the touchscreen monitor and microscope.
System	
Storage	32 GB DDR4 RAM; 512 GB NVMe PCIe SSD.
Output ports	Instrument: USB 3.1 Type B port; 4-pin power port. Computer: 1 USB 3.1 Gen 2 Type C port; 5 USB 3.1 Gen 1 Type A ports; 4 USB 2.0 Type A ports; 1 serial port; 2 display ports 1.2; 1 RJ45 port; 2 PS/2 ports; 1 UAJ port; 1 line out.
Networking capability	Connect via the Microsoft™ SMB protocol with an Ethernet cable or use the USB 3.0 WiFi dongle (included).
Cloud connectivity	Connect to the Thermo Fisher™ Connect Platform for remote access to images and data via a network connection.
Power supply	24 V AC adapter with country-specific power cords.
Physical characteristics	
Dimensions (W x D x H)	18 x 14 x 13 in. (45.7 x 33.0 x 35.6 cm)
Weight	26 lb (11.8 kg)

 Learn more at thermofisher.com/EVOSM7000

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